Lecture 6

Pulse Modulation

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Topics Covered (Chapter 7)

- Pulse Modulation
- Pulse Code Modulation
- Delta Converter
- Sigma-Delta Converter

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7-4: Pulse Modulation

- **Pulse modulation** is the process of changing a binary pulse signal to represent the information to be transmitted.
- The primary benefits of transmitting information by binary techniques are
  - Noise tolerance
  - Ability to regenerate a degraded signal.

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7-4: Pulse Modulation

- There are four basic forms of pulse modulation:
  1. Pulse-amplitude modulation (PAM)
  2. Pulse-width modulation (PWM)
  3. Pulse-position modulation (PPM)
  4. Pulse-code modulation (PCM).

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7-4: Pulse Modulation

Comparing Pulse-Modulation Methods

- The following slide shows an analog modulating signal and the various waveforms produced by PAM, PWM, and PPM modulators.
- In all three cases, the analog signal is sampled, as it would be in A/D conversion.

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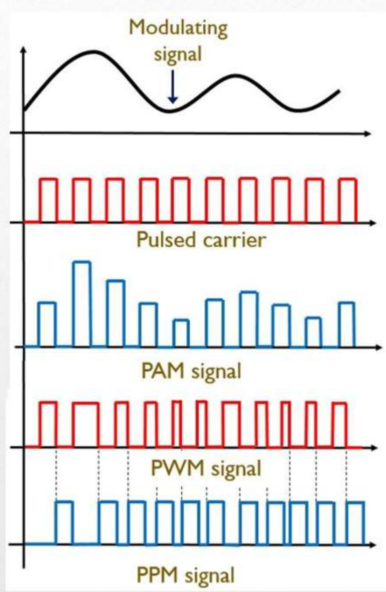

Types of pulse modulation.

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7-4: Pulse Modulation

Comparing Pulse-Modulation Methods

- The PAM signal is a series of constant-width pulses whose amplitudes vary in accordance with the analog signal.
- The PWM signal is binary in amplitude (has only two levels). The information signal varies the width or time duration of the pulse.
- In PPM, the pulses change position according to the amplitude of the analog signal.
- Of the four types of pulse modulation, PAM is the simplest and least expensive to implement.

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7-4: Pulse Modulation

Pulse-Code Modulation

- The most widely used technique for digitizing information signals for electronic data transmission is **pulse-code modulation (PCM).**
- PCM signals are serial digital data.
- There are two ways to generate:
  1. Use an S/H circuit and traditional A/D converter to sample and convert the analog signal into a sequence of parallel binary words, which are then converted into a serial form and transmitted serially.
  2. Use a delta modulator.

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7-4: Pulse Modulation

Pulse-Code Modulation: Traditional PCM

- In traditional PCM, the analog signal is sampled and converted into a sequence of parallel binary words by an A/D converter.
- The parallel binary output word is converted into a serial signal by a shift register.
- Each time a sample is taken, an 8-bit word is generated by the A/D converter.
- This word must be transmitted serially before another sample is taken and another word is generated.
- The clock and start conversion signals are synchronized so that the resulting output signal is a continuous train of binary words.

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7-4: Pulse Modulation

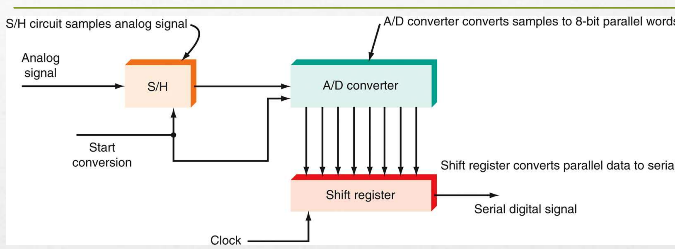

Figure 7-31: Basic PCM system.

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7-4: Pulse Modulation

Pulse-Code Modulation: Companding, Codecs/ Vocoders

- **Companding** is a process of signal compression and expansion that is used to overcome problems of distortion and noise in the transmission of audio signals.
- Companding is the most common means of overcoming the problems of quantizing error and noise.
- All A/D and D/A conversion and related functions, as well as companding, are taken care of by a single large-scale IC chip known as a **codec** or **vocoder**.

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A-Law Companding

- Companding: the process of *compressing* the analogue signal at the transmitter and *expanding* the signal at the receiver. The purpose of companding is to improve the SQNR. Uses DSP to implement companding and follows the equation:

$$y = \begin{cases} \dfrac{Ax}{1+\ln(A)}, & 0 \le x \le 1/A \\ \dfrac{1+\ln(Ax)}{1+\ln(A)}, & 1/A \le x \le 1 \\ -y(|x|), & -1 \le x \le 0 \end{cases}$$

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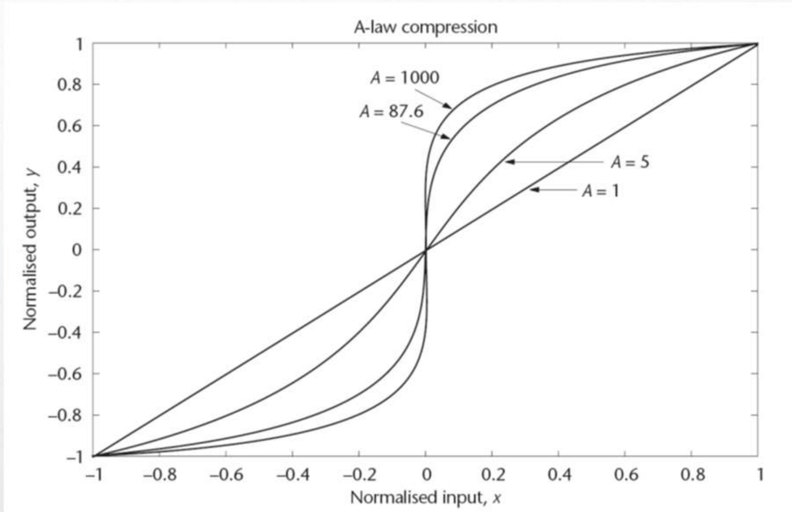

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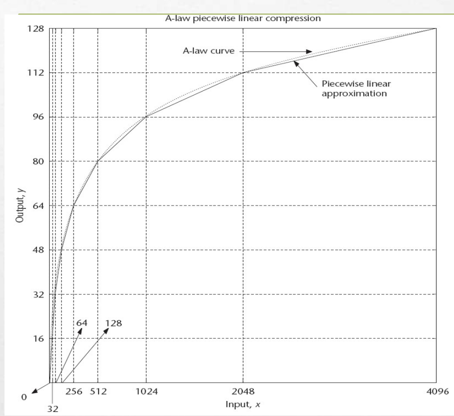
A-Law Companding



8 bit coding

MSB (b7): sign bit

1=positive, 0=negative

b6, b5, b4: segment number.

b3, b2, b1, b0: number of the interval within the segment. SNQR varies between 30–39 dB per segment. For the uniform equivalent, SQNR varies between 6–48 dB.

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Differential PCM

- When audio or video signal are sampled, it is usually found that adjacent samples are close in their values, i.e. there is a lot of redundancy resulting in a waste of bandwidth.
- DPCM transmitter predicts next sample based on previous samples, and transmits the quantised difference between actual and predicted next samples.
- At receiver, samples are regenerated by using previously reconstructed samples plus the quantised differential values received.
- Enables SQNR improvement as large as 25 dB compared to traditional PCM. Moreover, fewer bits can be used.

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Delta Modulation

- Delta modulation is a low-cost version of DPCM, where the codeword is 1 bit long: a "0" or a "1".
- Only the change of information: an increase or decrease of signal amplitude from the previous sample, is sent.
- Small or no amplitude change causes the modulated signal to oscillate between 0 or 1 state: **Granular noise**.
- If the input signal changes too rapidly to be tracked accurately, distortion in the output occurs: **Slope overload noise**.

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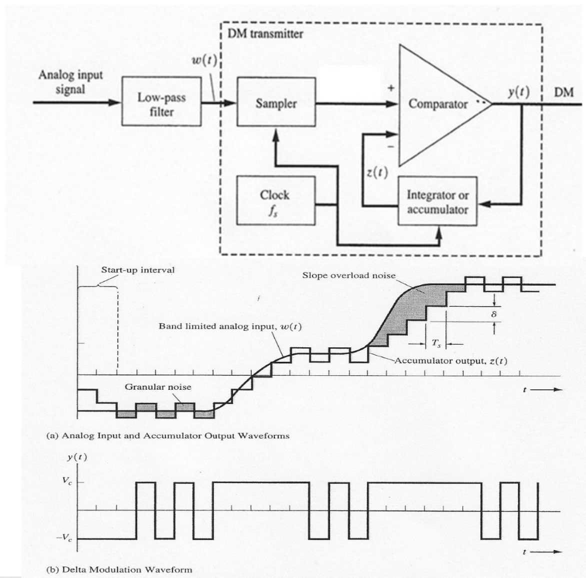

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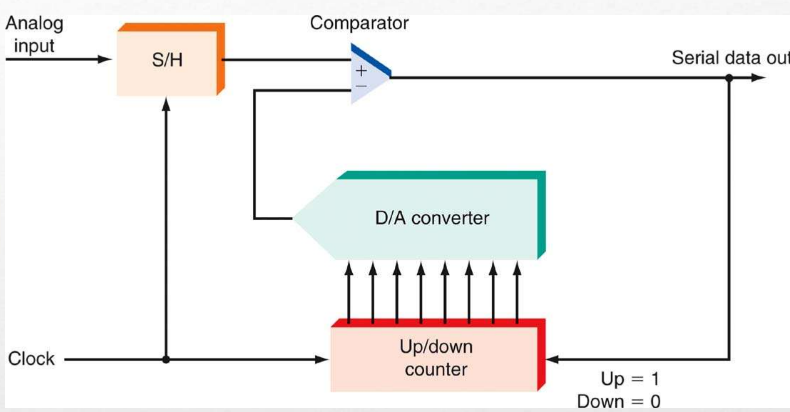

Figure 7-26: Delta modulator

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Delta Modulation

- The analog signal is sampled by an S/H circuit.
- The sample is also applied to a comparator.
- The other input to the comparator comes from a D/A converter driven by an up-down counter.
- The counter counts up (increments) or down (decrements) depending on the output state of the comparator.
- The comparator output is also the serial data signal representing the analog value.

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Sigma-Delta (ΣΔ) Converter

- A variation of delta converter. Also known as **delta-sigma** or **charge balance converter**.
- Offers extreme precision, wide dynamic range, and low noise.
- Available with word output lengths of 18, 20, 22, and 24 bits.
- These converters are widely used in digital audio applications (e.g. CD and MP3 players).

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Sigma-Delta Converter

- It uses a clock or sampling frequency that is many times the minimum Nyquist rate required, i.e. **oversampling**.
- This eliminates the problem of aliasing.
- Oversampling also spreads out quantization noise over wider frequency range, thus less noise within the actual signal range
- Feedback loop further pushes the noise into higher frequencies (noise shaping) that can be removed by a low-pass filter.

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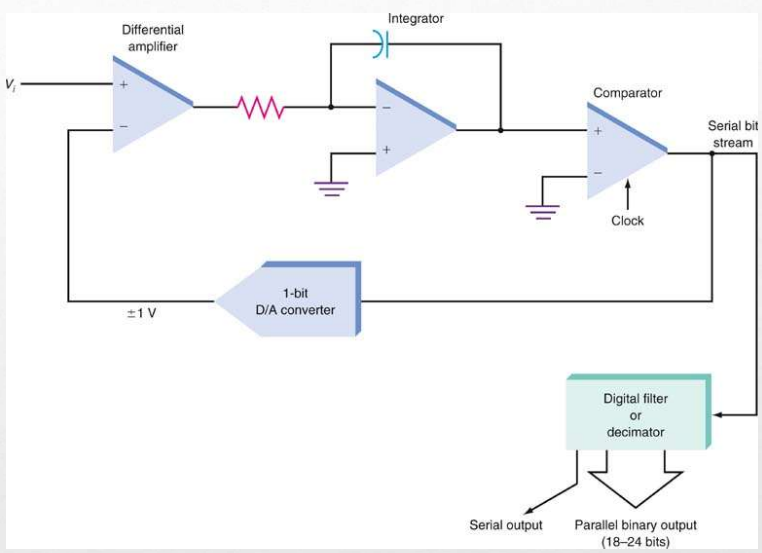

Fig. 7-29: A sigma-delta (ΣΔ) converter.

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